

History as Identity: The Append-Only Algebra of Spharepop

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Abstract

Spharepop is a computational framework whose defining departure from conventional models lies in its treatment of irreversibility as constitutive rather than incidental. Where standard semantic accounts collapse a computation to its observable output—discarding the path by which that output was reached—Spharepop preserves the ordered sequence of exclusions that produced the outcome and treats that sequence as the primary object. This essay develops the formal structure underlying this commitment. We introduce a minimal state representation consisting of a residual option space and an accumulated commitment history, define three primitive operators—POP, BIND, and COLLAPSE—and show that the resulting algebra is essentially a free history category whose morphisms are finite sequences of irreversible commitments rather than equivalence classes of traces. The contrast with traced monoidal categories is examined in detail: where the trace operator abstracts over internal computation by collapsing feedback into a fixed-point morphism, Spharepop refuses that identification and elevates the internal path to first-class status. The identity of a Spharepop process is therefore its causal history, not its input-output relation. We argue that this makes Spharepop the natural categorical semantics for systems in which provenance, auditability, and ordered commitment matter—systems whose mathematical structure is precisely that of event-sourced append-only logs.

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1 Introduction

A recurring ambition in the theory of computation is the identification of the minimal mathematical structure required to model a class of phenomena faithfully. For deterministic sequential computation that structure is the Turing machine or equivalently the λ -calculus; for concurrent interaction it is process algebra or the π -calculus; for feedback and recursion within monoidal categories it is the traced monoidal category of Joyal, Street, and Verity. In each case the theory achieves its power by deciding what to abstract over and what to preserve. Traces, in particular, achieve their power precisely by abstraction: the trace operator $\text{Tr}_{A,B}^U(f)$ for a morphism $f : A \otimes U \rightarrow B \otimes U$ asserts that the precise internal behavior on the “feedback wire” U is irrelevant once its effect on the external interface $A \rightarrow B$ is determined. The loop is collapsed; what remains is the fixed-point input-output relation.

Spherepop originates from the refusal of precisely this identification. Its central commitment is that the internal path—the ordered sequence of decisions, exclusions, and contextual restrictions that produced an observable state—is not auxiliary information to be discarded once a result is known. It is the result. Two computations that terminate at the same value but follow different causal chains are, in Spherepop’s ontology, distinct objects. The identity of a process is its history, not its output.

This essay develops the formal structure that underlies that commitment. Section 2 reviews traced monoidal categories and isolates precisely what the trace operator abstracts over. Section 3 introduces the Spherepop state representation as a pair consisting of a residual option space and an ordered commitment history. Section 4 defines the three primitive operators—POP, BIND, and COLLAPSE—and works through the minimal nontrivial example in detail. Section 6 characterizes the resulting algebra as a free history category and compares its structure to path objects in causal categories. Section 7 shows how apparent cyclic behavior emerges from dense chains of irreversible commitments. Section 8 draws connections to event sourcing, version control, and other domains where append-only provenance structures appear naturally.

2 The Trace Operator and What It Hides

Before examining what Spherepop preserves, it is useful to be precise about what the trace operator discards. Let $(\mathcal{C}, \otimes, I)$ be a symmetric monoidal category. Recall the definition.

Definition 2.1 (Traced monoidal category). *A traced symmetric monoidal category is a*

symmetric monoidal category $(\mathcal{C}, \otimes, I)$ equipped with a family of functions

$$\mathrm{Tr}_{A,B}^U : \mathcal{C}(A \otimes U, B \otimes U) \longrightarrow \mathcal{C}(A, B),$$

one for each triple of objects A, B, U , satisfying naturality, dinaturality, vanishing (the trace over I reduces to the identity), and the yanking equation $\mathrm{Tr}_{U,U}^U(\sigma_{U,U}) = \mathrm{id}_U$, where σ is the symmetry.

The informal reading is that a morphism $f : A \otimes U \rightarrow B \otimes U$ can be “looped” by feeding the U -component of its output back into its U -component input, yielding a morphism $A \rightarrow B$ that captures only the external behavior. The dinaturality condition ensures that the trace is insensitive to the choice of representative for the internal computation: any two morphisms $g : A \otimes U \rightarrow B \otimes U$ and $h : A \otimes V \rightarrow B \otimes V$ related by a change of the internal type $U \cong V$ in the appropriate sense produce equal traces.

This is a remarkable abstraction, and it is precisely the right one for a wide class of applications. In the semantics of recursive programs, the trace models fixed-point recursion: the loop corresponds to the recursive call, and the external morphism $A \rightarrow B$ captures the input-output behavior of the recursion without committing to a particular unrolling. In the geometry of interaction interpretation of linear logic, the trace models cut-elimination, and the internal computation on U corresponds to the steps of proof reduction that are erased once the normal form is reached. In each case the power of the trace comes from the fact that it treats the internal path as a *bureaucratic detail*—real and present during computation, but irrelevant to the mathematical object that results.

Spherepop makes the opposite architectural choice. What the trace treats as bureaucratic detail—the ordered sequence of internal steps—is precisely what Spherepop elevates to the role of identity. To understand why, it is necessary to examine what information is lost when a feedback loop is collapsed.

Remark 2.1. *Consider two programs P_1 and P_2 that both compute the identity function $A \rightarrow A$, but where P_1 reaches this output after zero recursive calls and P_2 reaches it after one thousand recursive calls that happen to cancel. In the traced monoidal semantics both programs denote the same morphism id_A . In Spherepop they denote distinct objects because the histories of commitments they produce differ. For systems in which the number, order, or content of intermediate commitments carries physical, informational, or legal significance, the trace semantics loses information that the system cannot afford to discard.*

This is not a defect of traced monoidal categories; it is a consequence of the problem they are designed to solve. The relevant question is whether the problem of interest is one in which

histories matter. Spherepop asserts that for a wide and important class of systems—those governed by irreversibility—they do.

3 State Representation: Option Space and History

The primitive data structure of Spherepop is a pair representing the current state of an unfolding process.

Definition 3.1 (Spherepop state). *Let Ω be a finite set called the option space. A Spherepop state is a pair*

$$(\Omega_t, H_t)$$

where $\Omega_t \subseteq \Omega$ is the residual option space at time t —the set of possibilities that have not yet been excluded or committed—and $H_t = [x_1, x_2, \dots, x_k]$ is the commitment history, an ordered list of elements that have been selected and recorded.

The initial state of any computation is $(\Omega, \emptyset)$: the full option space is available and no commitments have yet been made. As the computation proceeds, Ω_t shrinks monotonically (possibilities are excluded but never reintroduced) and H_t grows monotonically (commitments are appended but never retracted). These two monotonicity conditions are not incidental features of the notation but constitute the central structural constraint of the framework.

The asymmetry between Ω_t and H_t is deliberate and significant. The residual option space Ω_t carries no ordering: it is a set, recording which possibilities remain without recording when or in what sequence alternatives were ruled out. The history H_t , by contrast, is an ordered list: it records not merely which commitments were made but the sequence in which they occurred. This asymmetry reflects the asymmetry of time. The future is a space of undifferentiated possibilities; the past is an ordered causal chain.

Example 3.1. *Let $\Omega = \{a, b, c\}$. The initial state is $(\{a, b, c\}, \emptyset)$. After the commitment a , the state becomes $(\{b, c\}, [a])$. After a subsequent contextual restriction that eliminates c without committing to it, the state becomes $(\{b\}, [a])$. After the final commitment b , the state becomes $(\emptyset, [a, b])$.*

Two alternative computations over the same option space might produce the states $(\emptyset, [a, b])$ and $(\emptyset, [b, a])$. Both have exhausted the option space, and if the observable outcome is merely the fact of exhaustion, the two computations appear equivalent. In Spherepop they are not: the histories $[a, b]$ and $[b, a]$ are distinct ordered lists, and the identity of each process is its history.

The state representation of Definition 3.1 is minimal in the following sense: it records exactly the information needed to reconstruct the observable state of the system via the COLLAPSE operator defined in Section 4, and no more. The discarded alternatives in $\Omega \setminus \Omega_t$ are genuinely lost to the history; they do not appear in H_t because they were excluded rather than committed.

4 The Three Primitive Operators

Three operators act on Spherepop states. Together they constitute the complete operational vocabulary of the framework.

4.1 Pop: Irreversible Commitment

POP is the fundamental operator of Spherepop. It performs an irreversible selection from the current residual option space and records that selection in the history.

Definition 4.1 (POP). *Let (Ω_t, H_t) be a Spherepop state and let $x \in \Omega_t$. The operator Pop_x acts as*

$$\text{Pop}_x(\Omega_t, H_t) = (\Omega_t \setminus \{x\}, H_t \cdot x),$$

where $H_t \cdot x$ denotes the list obtained by appending x to H_t .

The key properties of POP are the following. First, x is removed from Ω_t : once an element is committed, it is no longer available as a future possibility. Second, x is appended to H_t : the commitment is recorded in causal order. Third, the operation is irreversible: there is no operator Pop^{-1} that restores x to the option space and removes it from the history. The option space strictly shrinks with each application.

This irreversibility is the mathematical expression of a fact about the systems Spherepop is designed to model. In any process governed by genuine commitment—the signing of a contract, the emission of a photon, the writing of a git commit, the irreversible bifurcation of a developmental pathway—the committed state cannot be undone. The alternatives that were not chosen at the moment of commitment do not persist as latent possibilities; they are foreclosed.

Remark 4.1. *The strict decrease in $|\Omega_t|$ after each POP implies that a computation over a finite option space Ω terminates after at most $|\Omega|$ applications of POP. The computation is therefore guaranteed to be finite provided the option space is finite and no BIND operation introduces new elements. Spherepop is not designed for infinite option spaces without modification; the finite case already exhibits all the structural features of interest.*

4.2 Bind: Contextual Restriction

BIND performs a contextual restriction of the option space without making a commitment. It applies an admissibility condition that narrows the set of future possibilities without recording any element in the history.

Definition 4.2 (BIND). *Let $f : 2^\Omega \rightarrow 2^\Omega$ be a function satisfying $f(\Omega_t) \subseteq \Omega_t$ for all Ω_t (that is, f is an interior operator on the powerset lattice). The operator Bind_f acts as*

$$\text{Bind}_f(\Omega_t, H_t) = (f(\Omega_t), H_t).$$

The history H_t is unchanged by BIND: because no commitment has been made, there is nothing to record. The option space Ω_t is replaced by $f(\Omega_t)$, which is smaller or equal. The elements in $\Omega_t \setminus f(\Omega_t)$ are silently excluded—they are not committed and therefore do not appear in H_t , but they are no longer available for future POP operations.

This silent exclusion is a philosophically significant feature. BIND models the operation of a contextual constraint: a precondition, an admissibility filter, a physical law that rules out certain trajectories before any selection occurs. The excluded elements leave no trace in the causal history because they were never genuinely in play at the moment of commitment; they were ruled out by the structure of the problem rather than by a choice made within it.

Example 4.1. *In the option space $\Omega = \{a, b, c\}$, the function $f(\Omega_t) = \Omega_t \setminus \{c\}$ represents the constraint “ c is inadmissible.” Applying Bind_f to the initial state:*

$$\text{Bind}_f(\{a, b, c\}, \emptyset) = (\{a, b\}, \emptyset).$$

The element c has been excluded, but the history remains empty. The subsequent computation proceeds as if c never existed as a genuine option.

The distinction between exclusion by BIND and exclusion by POP corresponds to the distinction between structural impossibility and chosen commitment. An element excluded by BIND was never a real candidate; an element excluded by POP was a genuine alternative that the process passed over. This distinction may or may not be recoverable from an external observation of the final state, but it is always present in the SpheroPOP state pair.

4.3 Collapse: Reconstruction from History

COLLAPSE is the inverse operation in a loose sense: it reconstructs an observable state from the accumulated commitment history. It is not an inverse of POP in the categorical sense—it does not restore the option space—but rather a functor from histories to observable states.

Definition 4.3 (COLLAPSE). *Let s_0 be a fixed root state and let $\{T_x\}_{x \in \Omega}$ be a family of state transformations, one for each element of the option space. For a history $H_t = [x_1, x_2, \dots, x_n]$, the operator Collapse acts as*

$$\text{Collapse}(H_t) = T_{x_n} \circ T_{x_{n-1}} \circ \dots \circ T_{x_1}(s_0).$$

The observable state is not primitive. It is derived from the sequence of commitments by composing the associated transformations from left to right in historical order. The earliest commitment x_1 acts first on the root state s_0 , and the most recent commitment x_n acts last. The result is the state that the system would be in if it had actually executed those commitments in that sequence.

This derivation structure is the deepest claim of the Spherpap ontology. States are not given; they are constructed. What is given is the root state s_0 —which may be thought of as the vacuum or the origin—and the ordered list of commitments. All other states are derived objects, computable from the history by COLLAPSE.

Remark 4.2. *The composition order in COLLAPSE is that of diagrammatic composition in a category: T_{x_1} is applied first, T_{x_n} last. This matches the temporal order of commitments. Note that in general $T_{x_i} \circ T_{x_j} \neq T_{x_j} \circ T_{x_i}$: the history $[a, b]$ and the history $[b, a]$ may yield different observable states via COLLAPSE, even though they contain the same set of elements. Commutativity of the transformations is an additional assumption that must be imposed if and when it holds; it is not part of the definition.*

5 The Minimal Example: Three Operators over $\{a, b, c\}$

Having defined the three operators, it is instructive to trace through the minimal nontrivial Spherpap computation in full detail. Let $\Omega = \{a, b, c\}$, let the initial state be $(\{a, b, c\}, \emptyset)$, and let s_0 be the root state.

Step 0. The computation begins:

$$(\{a, b, c\}, \emptyset).$$

All three possibilities are available and no commitments have been made.

Step 1: Apply Bind_f with $f(\Omega_t) = \Omega_t \setminus \{c\}$. The constraint “ c is inadmissible” narrows the option space:

$$(\{a, b\}, \emptyset).$$

The history remains empty because no commitment has occurred. The element c has been silently excluded by context.

Step 2: Apply Pop_a . The system commits to a :

$$(\{b\}, [a]).$$

The option space now contains only b . The history records that a was the first commitment. The identity of the process, at this moment, is the ordered sequence $[a]$.

Step 3: Apply $\text{Collapse}([a])$. The observable state is reconstructed:

$$s = T_a(s_0).$$

Even though the residual option space still contains b , the observable state of the system is $T_a(s_0)$ —the root state transformed by the single committed element a .

This minimal computation already exposes the central structural difference between Spherpap and conventional evaluation. A traditional functional model would compute the value $T_a(s_0)$ and return it, discarding the alternative b and the contextual exclusion of c as implementation details. In Spherpap the pair $(\{b\}, [a])$ is the genuine object of study. The residual alternative b remains present in the state, recording that the computation was made in a context where b was a genuine possibility that was passed over. The exclusion of c is likewise recorded—albeit silently, in the gap between Ω and Ω_t —as a constraint of the environment rather than a choice of the process.

Observation on identity. Two Spherpap computations over $\Omega = \{a, b, c\}$ that produce the same observable state $s = T_a(s_0)$ may differ in their causal histories. A computation that applies Bind_f before Pop_a produces the state $(\{b\}, [a])$. A computation that applies only Pop_a with no prior BIND produces the state $(\{b, c\}, [a])$. Both yield the same observable state via COLLAPSE . In a purely extensional semantics they are the same. In Spherpap they are distinct: the first computation occurred in a context where c had already been ruled out; the second occurred in a context where c remained a live possibility. The causal provenance of the commitment differs, and that difference is preserved in the state representation.

6 The History Algebra and the Free History Category

The three operators of Section 4, together with the state representation of Section 3, constitute what we call the *history algebra* of Spheredpop. We now characterize its mathematical structure more precisely.

Definition 6.1 (History algebra). *A history algebra over a finite option space Ω consists of:*

- (i) *the set of all Spheredpop states $\{(\Omega_t, H_t) \mid \Omega_t \subseteq \Omega, H_t \in \Omega^*\}$, where Ω^* denotes the set of finite ordered lists over Ω ;*
- (ii) *for each $x \in \Omega$, the operator Pop_x acting on states with $x \in \Omega_t$;*
- (iii) *for each admissibility function f , the operator Bind_f ;*
- (iv) *a root state s_0 and a family $\{T_x\}_{x \in \Omega}$ of transformations defining the operator Collapse.*

The algebraic laws governing these operators are sparse but decisive. POP operations compose strictly by concatenation: $\text{Pop}_{x_2} \circ \text{Pop}_{x_1}$ produces history $[x_1, x_2]$, not $[x_2, x_1]$. The composition is therefore non-commutative in general. BIND operations compose by function composition: $\text{Bind}_{f_2} \circ \text{Bind}_{f_1}$ applies f_1 first and f_2 second. POP and BIND interact as follows: $\text{Pop}_x \circ \text{Bind}_f$ is defined only when $x \in f(\Omega_t)$, that is, only when the admissibility filter has not excluded x before the commitment.

The composition structure of histories is that of a free monoid Ω^* under concatenation, with the empty history $\emptyset$ as the identity element. This is the universal algebraic object generated by $|\Omega|$ generators with no relations—the freest possible monoid, with no equations identifying distinct words. That there are no relations is not a deficiency; it is the point. The history algebra imposes no equivalences on distinct commitment sequences, just as the free monoid imposes no equations on distinct words. Two different histories are always different objects.

This observation motivates the categorical characterization of Spheredpop.

Definition 6.2 (Free history category). *The free history category $\mathcal{H}(\Omega)$ over a finite option space Ω is the category whose:*

- (i) *objects are pairs (Ω_t, H) where $\Omega_t \subseteq \Omega$ and $H \in \Omega^*$;*
- (ii) *morphisms from (Ω_t, H) to (Ω'_t, H') are pairs (σ, τ) where $\sigma : \Omega_t \rightarrow \Omega'_t$ is a set map and $\tau \in \Omega^*$ satisfies $H' = H \cdot \tau$ —that is, H' extends H by the commitment sequence τ ;*

(iii) composition is given by $(\sigma', \tau') \circ (\sigma, \tau) = (\sigma' \circ \sigma, \tau \cdot \tau')$;

(iv) identity morphisms are $(\text{id}_{\Omega_t}, \emptyset)$ —the identity commits nothing and leaves the option space unchanged.

Proposition 6.1. *The category $\mathcal{H}(\Omega)$ is a well-defined category. The POP and BIND operators of Section 4 correspond to generating morphisms in $\mathcal{H}(\Omega)$, and COLLAPSE is a functor $\text{Collapse} : \mathcal{H}(\Omega) \rightarrow \mathcal{S}$ where $\mathcal{S}$ is the category of observable states and transformations.*

Proof. Associativity of composition follows from associativity of concatenation in Ω^* and associativity of function composition. Identity morphisms satisfy $(\sigma', \tau') \circ (\text{id}, \emptyset) = (\sigma', \tau')$ and $(\text{id}, \emptyset) \circ (\sigma, \tau) = (\sigma, \tau)$, as required. The functor Collapse sends a history $H = [x_1, \dots, x_n]$ to the composite transformation $T_{x_n} \circ \dots \circ T_{x_1}$; functoriality follows from the fact that concatenation of histories corresponds to composition of transformations. $\square$

The contrast with traced monoidal categories can now be stated precisely. In a traced monoidal category $\mathcal{C}$, two morphisms $f, g : A \otimes U \rightarrow B \otimes U$ that produce the same trace $\text{Tr}_{A,B}^U(f) = \text{Tr}_{A,B}^U(g) : A \rightarrow B$ are identified for the purposes of the external interface. The trace is a quotient: it identifies morphisms whose internal behavior differs but whose external effect is the same. In $\mathcal{H}(\Omega)$ there is no such quotient. Two morphisms with the same source, target, and observable output but different commitment sequences $\tau \neq \tau'$ are distinct morphisms. The category $\mathcal{H}(\Omega)$ is not a quotient of anything; it is a free object. Its morphisms are literally the words in the free monoid Ω^* , with no equivalences imposed.

Remark 6.1 (Path objects and causal categories). *The morphisms of $\mathcal{H}(\Omega)$ are path objects in the sense of abstract homotopy theory: they are not equivalence classes of paths but the paths themselves, with path identity being literal equality of ordered sequences. This is also the sense in which a Spherpap computation is a path object: its identity is not the homotopy class of its trajectory through configuration space but the specific ordered sequence of events that constituted it. The categorical semantics of Spherpap is therefore not that of a quotient category but that of a path groupoid—or rather, since the paths are irreversible, a path category with no inverses.*

7 Apparent Cycles from Dense Irreversible Chains

One of the most counterintuitive features of the Spherpap model is the way cyclic behavior emerges from what is, at the level of internal structure, a strictly non-cyclic process. The option space shrinks monotonically and the history grows monotonically; there is no literal

loop. Yet external observers of a Spherpap computation frequently perceive periodicity, recurrence, or feedback. This section explains the mechanism.

Consider a computation that proceeds as a long alternating sequence of BIND and POP operations:

$$\text{Bind}_{f_1} \rightarrow \text{Pop}_{x_1} \rightarrow \text{Bind}_{f_2} \rightarrow \text{Pop}_{x_2} \rightarrow \cdots \rightarrow \text{Bind}_{f_n} \rightarrow \text{Pop}_{x_n}.$$

The resulting history is $H = [x_1, x_2, \dots, x_n]$. Suppose further that the admissibility functions $\{f_k\}$ are *renewing*: at each stage k , the function f_k restricts the current option space to a subset that resembles the option space available at some earlier stage $k' < k$. For instance, if $\Omega = \{a, b, c\}$ and the computation alternately commits to a and then reintroduces a fresh copy of $\{a, b, c\}$ through a BIND operation that acts on an extended option space, the external observer—who sees only the sequence of committed elements—may observe the pattern $[a, a, a, \dots]$ and infer a cycle.

The inference is correct at the macroscopic level and incorrect at the structural level. At the macroscopic level, the external output of the computation does exhibit periodicity. At the structural level, the history $[a, a, a, \dots]$ is a strictly growing list. Each occurrence of a is a distinct commitment at a distinct moment in the causal chain. The apparent cycle is the macroscopic shadow of a rapidly extending causal thread.

Proposition 7.1 (Apparent cycles from dense irreversible chains). *Let Ω be a finite option space and let $\{\Omega^{(k)}\}$ be a sequence of option spaces with $\Omega^{(k+1)} \supseteq \Omega^{(k)}$ for all k (i.e., the effective option space is periodically refreshed by context). Let a Spherpap computation commit to elements $x_1, x_2, x_3, \dots$ from successive refreshed spaces. If the commitment rule is periodic in the sense that $x_{k+p} = x_k$ for some period p and for all $k \geq k_0$, then the observable output sequence via COLLAPSE is periodic with period p , while the history $H = [x_1, x_2, \dots]$ is strictly non-repeating as a list.*

Proof. Periodicity of the output sequence follows by definition: $\text{Collapse}([x_1, \dots, x_n]) = T_{x_n} \circ \cdots \circ T_{x_1}(s_0)$, and if the committed elements repeat with period p , the composition of transformations repeats with the same period in its suffix. Non-repetition of the history follows from the fact that the history is an append-only list and H_{t+1} strictly contains H_t as a proper prefix. $\square$

The practical implication is that Spherpap can model cyclic phenomena—heartbeats, oscillating chemical reactions, periodic regulatory feedback, circadian rhythms—without introducing any cyclic structure into its underlying representation. The cycles are macroscopic invariants of a monotonically growing history. This is not merely a formal curiosity; it is pre-

cisely the structure of real physical and biological systems in which apparent cycles are the observable consequence of an underlying time-directed causal process that never genuinely repeats.

The contrast with traced monoidal categories is again instructive. In the traced setting, a cyclic computation is modeled by an actual loop: the morphism $f : A \otimes U \rightarrow B \otimes U$ has a genuinely cyclic feedback wire U , and the trace collapses the loop into a fixed-point. The cycle is in the morphism. In Spheredpop, the cycle is nowhere in the morphism; it is only in the external observation of the output sequence. The internal structure is always acyclic.

8 Connections to Append-Only Systems

The mathematical structure identified in the preceding sections—an append-only history, a shrinking option space, and an observable state derived by composing committed transformations—appears in several domains of computer science and distributed systems under different names. Recognizing these connections is not merely an observation of analogy; it suggests that Spheredpop is the natural abstract algebraic characterization of a pattern that arises wherever irreversibility and ordered provenance are structurally required.

Event sourcing. In the event sourcing architectural pattern, the state of a system is not stored directly but derived from an append-only log of events. Each event is immutable once recorded; the current state is reconstructed at any moment by replaying the event log from the beginning. This is precisely the structure of Spheredpop’s COLLAPSE operator: the observable state is $T_{x_n} \circ \dots \circ T_{x_1}(s_0)$, where each x_i corresponds to an event and s_0 is the initial state of the aggregate. The event log is the history H_t , and the domain aggregate is the result of COLLAPSE. The Spheredpop framework can therefore be understood as the categorical semantics of event-sourced systems, making precise the algebraic laws that such systems implicitly obey.

Version control and git. A git repository is an append-only directed acyclic graph of commits. Each commit records a transformation of the repository state relative to its parents, and the current state of any branch is derived by traversing the commit graph from the root. The history $H_t = [x_1, \dots, x_n]$ in Spheredpop corresponds to the sequence of commits on a linear branch; POP corresponds to the act of committing; BIND corresponds to the application of a merge filter or a pre-commit hook that restricts the admissible changes; and COLLAPSE corresponds to checking out—reconstructing the working tree state from the commit history. The non-commutativity of Spheredpop histories corresponds to the well-

known non-commutativity of git merges: rebasing a branch as $[a, b]$ produces a different repository state than rebasing it as $[b, a]$ whenever the transformations T_a and T_b modify overlapping files.

Ledgers and blockchains. A financial ledger is an append-only sequence of transactions. The current balance of any account is derived from the ledger by composing the signed transformations corresponding to each transaction—credits and debits applied in temporal order. A blockchain generalizes this by making the append-only property cryptographically enforced: each block references the hash of its predecessor, making retrospective alteration of the history computationally infeasible. The Spherepop framework provides a clean algebraic characterization of the invariants that both systems rely upon: the free history category $\mathcal{H}(\Omega)$ is the abstract algebra of a system whose identity is its ordered record of commitments.

Causal histories in physics. In the causal set approach to quantum gravity, spacetime is modeled as a locally finite partial order of events, where the partial order encodes the causal precedence relation. The history of a physical system is the causal set of events that led to its current state, and the state is derived by composing the effects of those events in causal order. This is formally identical to the COLLAPSE operator applied to a history H_t where the ordering is causal rather than merely temporal. The Spherepop framework may therefore be understood as a discretized version of causal set dynamics, with the option space Ω playing the role of the future light cone of admissible next events and each POP operation corresponding to the actualization of one event from that cone.

9 Conclusion

The foregoing analysis has developed the formal structure of Spherepop as a minimal append-only history algebra equipped with three operations: POP, which performs irreversible commitment; BIND, which applies contextual restriction without commitment; and COLLAPSE, which derives the observable state from the ordered commitment history. The natural categorical home of this algebra is the free history category $\mathcal{H}(\Omega)$, whose morphisms are ordered sequences of commitments in the free monoid Ω^* and whose composition law is list concatenation.

The structural contrast with traced monoidal categories is precise and illuminating. The trace operator achieves its power by identifying morphisms with the same external behavior, regardless of internal path—it is a quotient construction that abstracts over the details of feedback loops. The free history category $\mathcal{H}(\Omega)$ is the opposite extreme: it is the freest

category consistent with the monotonicity constraints of the Spherepop state, and it identifies no two distinct histories. The identity of a Spherepop process is its history, not its output. The trace collapses the path into a morphism; Spherepop refuses that collapse and keeps the path.

This refusal is not a deficiency. It is the correct formal response to a genuine structural requirement: the requirement that the provenance, order, and identity of intermediate commitments be preserved as first-class mathematical structure. The systems that satisfy this requirement—event-sourced architectures, version-controlled repositories, cryptographic ledgers, causal set models of spacetime—are not exotic edge cases. They are among the most consequential computational and physical systems that exist. Spherepop provides them with the categorical semantics they deserve.